

## Technology Stack

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| <b>Date</b>          | 15 JUNE 2026                                               |
| <b>Team ID</b>       | xxxxxx                                                     |
| <b>Project Name</b>  | Sustainable Waste Management Assistant Using Generative AI |
| <b>Maximum Marks</b> | 2 Marks                                                    |

| S.No | Architecture Component / Layer                       | Technology Chosen |                         | Justification / Purpose                                                                                                                                          |
|------|------------------------------------------------------|-------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side<br><br>(e.g., React, Angular, |                   | React.js + Tailwind CSS | React provides a responsive, component-based UI for fast updates and reusability; Tailwind CSS ensures modern, mobile-friendly design with minimal code overhead |

|   |                                                                         |                                                                                                                                                  |
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| 2 | <b>Backend / Server-Side</b><br><br>(e.g. Node.js, Python/Django, Java) | <i>Flask (Python)]</i><br><br><i>Lightweight and easy to integrate with AI models and APIs; ideal for RESTful services and rapid development</i> |
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|---|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 3 | <b>Dat<br/>abas<br/>e /<br/>Dat<br/>a<br/>Stor<br/>age</b><br><br>(e.g.<br>,<br>MyS<br>QL,<br>Mon<br>goD<br>B,<br>Post<br>greS<br>QL) | <i>Firebase Firesto<br/>re (Cloud NoSQ<br/>L)]</i>                         | <i>Real-time data sync for scan history and user records; scalabl<br/>e and serverless for low maintenance</i>     |
| 4 | <b>Clo<br/>ud /<br/>Hos<br/>ting<br/>/<br/>Dep<br/>loy<br/>men<br/>t</b>                                                              | <i>Firebase Hostin<br/>g + Local Flask<br/>Server (Vite for<br/>React)</i> | <i>Firebase offers secure, cost-effective hosting and easy deploy<br/>ment; Flask runs locally for API testing</i> |

|   |                                                                    |                              |                                                                                                                                                                                                              |
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|   | (e.g.<br>,<br>AWS<br>,<br>Azur<br>e,<br>Her<br>oku,<br>Doc<br>ker) |                              |                                                                                                                                                                                                              |
| 5 | <b>Ver<br/>sion<br/>Con<br/>trol<br/>&amp;<br/>CI/<br/>CD</b>      | <i>GitHub + Git CL<br/>I</i> | <i>Enables team collaboration, version tracking, and continuous<br/>integration for updates and bug fixes</i>                                                                                                |
|   | (e.g.<br>,<br>Git<br>Hub,<br>GitL<br>ab,<br>Jenk<br>ins)           |                              | Groq provides AI-powered waste classification and eco recom<br>mendations; Leaflet and OpenStreetMap enable free map integ<br>ration without paid keys; Chart.js creates interactive analytics<br>dashboards |

|   |                                                                                  |                                                                                                                                                                         |                                                                    |
|---|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 6 | <b>Third-Party APIs / Other Tools</b><br><br>(e.g., Stripe, Twilio, Google Maps) | <i>Groq API (LLaMA 3.3-70B Versatile) – AI classification and guidance</i><br><i>Leaflet.js + OpenStreetMap – map rendering</i><br><i>Chart.js – data visualization</i> | <i>[Explain the specific feature this enables in your project]</i> |
|---|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|